Stakeholder Report: Steering Wheel Repair Data Analysis

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Executive Summary

This report presents findings from Tasks 1–3 of the Axion Ray assignment:

- **Task 1:** Data validation and cleaning of repair records.
- **Task 2:** Preparation and integration of repair and order datasets.
- **Task 3:** Exploratory Data Analysis (EDA) to identify trends and root causes.

The analysis highlights recurring steering wheel failures, cost drivers, and integration challenges, with actionable recommendations for stakeholders.

1 Task 1: Data Validation & Cleaning

1.1 Column Analysis

Key fields include VIN, TRANSACTION_ID, CUSTOMER_VERBATIM, CAUSAL_PART_NM, REPAIR_DATE, and cost metrics.

1.2 Cleaning Summary

- Removed rows missing VIN or transaction IDs.
- Standardized categorical fields (STATE, BODY_STYLE).
- Converted cost fields to numeric and capped extreme outliers.
- Deduplicated VIN + TRANSACTION_ID combinations.

1.3 Tags Generated

From free-text complaints, tags were extracted:

- Heated Steering Wheel Failure
- Horn Inoperative
- Trim/Applique Issue
- Leather/Stitching Defect

2 Task 2: Data Preparation & Integration

2.1 Primary Key

VIN (Task 1) $\leftrightarrow$ SERIAL NUMBER (Task 2) chosen as integration key. Fallback: TRANSACTION_ID $\leftrightarrow$ PRIMARY KEY.

2.2 Integration Approach

- Left Join used to preserve all Task 1 records.
- Justification: ensures continuity for repair analytics.
- Alternative: Inner Join would restrict analysis to overlapping records, risking data loss.

3 Task 3: Exploratory Data Analysis (EDA)

3.1 Trends Identified

1. Repair Volume Over Time – peaks suggest seasonal or batch quality issues.
2. Cost Correlation – strong correlation between labor cost and total cost.
3. Failure Frequency by Component – heated steering wheel modules and leather/stitching defects most frequent.

3.2 Root Cause Insights

- Heated Steering Wheel Modules – High frequency and cost → design flaw requiring redesign.
- Leather/Stitching Defects – Recurring cosmetic issue → supplier quality problem.
- Horn Harness Failures – Electrical connection issues → assembly defect.

4 Key Takeaways & Recommendations

- Discrepancies: Missing VINs, inconsistent categorical values, extreme cost outliers addressed during cleaning.
- Tags Summary: Heated steering wheel failures and leather/stitching defects are recurring, high-impact issues.
- Recommendations:
  - Investigate supplier quality for steering wheel components.
  - Redesign heated steering wheel modules.
  - Monitor regional variation in defect rates.
- Additional Observations:
  - Colder climates show higher heated steering wheel failures.
  - Cosmetic defects are widespread but lower cost compared to electrical failures.